

RippleLab Day 10 Progress Report

Day: 10 of 30

Date: 2026-08-15

Status: Complete

Branch: codex/day-10-oil-price

Implementation commit: 95fa368

Deployed commit: 95fa368

Production version: 6

Quality gate: Full local suite passed

1. Objective

Build RippleLab's personal oil-price engine. Day 10 must keep a global crude-oil scenario separate from local retail fuel, calculate direct vehicle-fuel exposure and explicit indirect effects on transport, food and utilities, show why different household exposure patterns produce different results, and return the same inspectable `SimulationResult` envelope used by the earlier scenarios.

2. Acceptance Criteria

- [x] Add a canonical `oil_price_change` request example.
- [x] Distinguish crude oil in USD per barrel from retail fuel in paise per litre.
- [x] Calculate direct fuel impact from entered monthly litres.
- [x] Calculate indirect transport, food and utility effects.
- [x] Keep all four transmission assumptions visible and editable.
- [x] Seed household expenses from the saved financial profile.
- [x] Add commuter, transit-renter and low-driving exposure presets.
- [x] Return impacts, sensitivity, assumptions, evidence, confidence and warnings.
- [x] Return a seven-node, nine-edge inspectable causal graph.
- [x] Add a protected Next.js proxy and FastAPI simulation endpoint.
- [x] Cover API, rising/falling oil, low-driving, validation, desktop, mobile and accessibility cases.
- [x] Update GitHub, the PDF progress record and the owner-only production deployment.

3. Delivered User Experience

A signed-in user opens the Oil-price simulator from the dashboard or navigation. RippleLab starts with the saved expense profile, offers three transparent exposure presets and allows every value to be edited before calculation.

```
Current and target crude price in USD/barrel
-> editable crude-to-retail transmission
-> modeled retail fuel price in INR/litre
-> direct fuel effect from monthly litres
-> indirect transport, food and utility effects
-> annual household total and sensitivity range
```

The output separates the target crude scenario, modeled retail fuel price, direct annual fuel effect and combined indirect effect. A driver table preserves the four household channels, while an inspectable causal graph exposes every mechanism, formula, assumption, source, lag and confidence score.

4. Household Exposure Design

The saved profile does not yet contain detailed fuel consumption or category ledgers. Day 10 therefore creates visible starting inputs rather than inferring hidden precision.

Input	Starting value for seeded profile	Purpose
Monthly fuel use	40 litres	Direct vehicle-fuel exposure
Transport services	20% of essential expenses	Non-fuel transport exposure
Food and groceries	40% of essential expenses	Indirect food exposure
Utilities	20% of essential expenses	Indirect household-energy exposure
Current retail fuel	INR 100 per litre	Editable local starting point

The Car commuter preset retains the profile-derived starting point. Transit renter reduces direct fuel to 8 litres and transport spending to 60% of the starting allocation. Low-driving reduces direct fuel to 5 litres and transport spending to 35%. Presets update form inputs only and never overwrite the saved financial profile.

5. Deterministic Model

Crude-price change

```
crude change = (target crude - current crude) / current crude
```

Scenario formula: `scenario.oil_price_change.v1`

Crude to retail fuel

```
target retail price = current retail price x (1 + crude change x retail pass-through)
```

Formula version: `oil.crude_to_retail.v1`

Direct vehicle fuel

```
annual effect = -monthly litres x (target retail price - current retail price) x 12
```

Formula version: `oil.direct_fuel_cost.v1`

Indirect household expense

```
annual effect = -monthly spend x crude change x category pass-through x 12
```

Formula version: `oil.indirect_expense.v1`

The engine uses Python Decimal arithmetic and rounds money to integer paise with `ROUND_HALF_UP`. Positive values are household benefits and negative values are household costs. An LLM does not calculate money or causal totals.

6. Known-value Result

The seeded Bengaluru household uses a current crude scenario of USD 80 per barrel and target of USD 120 per barrel, a 50% increase. With an INR 100 retail starting price and 30% crude-to-retail pass-through, the modeled retail price is INR 115 per litre.

Output	Verified value
Modeled retail fuel price	INR 115.00 per litre
Direct fuel annual effect	-INR 7,200.00
Transport annual effect	-INR 7,920.00
Food annual effect	-INR 5,280.00
Utilities annual effect	-INR 3,960.00
Total annual household effect	-INR 24,360.00

Lowering monthly fuel use from 40 litres to 5 litres, while leaving indirect expenses unchanged, changes the annual result to -INR 18,060.00. A crude decrease from USD 80 to USD 40 produces a symmetric INR 24,360.00 modeled benefit under the same assumptions.

7. Exposure Preset Comparison

The three presets make group differences explainable through visible inputs rather than fixed demographic labels.

Exposure preset	Direct litres	Transport spend	Verified annual effect
Car commuter	40 litres	INR 11,000	-INR 24,360.00
Transit renter	8 litres	INR 6,600	-INR 15,432.00
Low-driving	5 litres	INR 3,850	-INR 12,912.00

Food and utility inputs remain constant in this comparison. The renter preset does not model rent changes; it represents low direct fuel use and public-transport exposure. This boundary is explicit so the product does not imply that housing tenure alone determines oil-price sensitivity.

8. Sensitivity and Confidence

The uncertainty range is a deterministic sensitivity check. The engine lowers every pass-through by 20 percentage points, recalculates the total, then raises every pass-through by 20 percentage points and recalculates again. Values are clamped between 0% and 200%.

For the seeded commuter scenario, the selected result is -INR 24,360.00. The lower-transmission result is -INR 2,400.00 and the higher-transmission result is -INR 81,960.00. The shared contract retains p10, p50 and p90 field names, but the method is `deterministic_bounds`, not a probability distribution.

Overall confidence is medium at 70/100. Household spending and fuel-use inputs are personalized and arithmetic is deterministic, while crude-to-retail and indirect consumer-price transmission remain uncertain scenario choices. Direct fuel confidence is 75/100 and indirect-channel confidence is 65/100.

9. Evidence

Day 10 uses primary official Indian sources:

- Petroleum Planning and Analysis Cell, [International Prices of Crude Oil \(Indian Basket\)](#) - official crude-price series and unit context.
- Petroleum Planning and Analysis Cell, [Price Build Up of Petrol and Diesel](#) - retail selling-price components.
- Ministry of Statistics and Programme Implementation, [Consumer Price Index frequently asked questions](#) - food, fuel and transport consumption-group context.

The sources support the economic categories and price context. They do not validate the selected pass-through values. No current crude or retail price is hard-coded as an observed fact; both are explicit user-editable scenario inputs.

10. Causal Graph

The engine returns seven nodes and nine edges:

```
Indian Basket crude scenario
-> Modeled retail fuel price -> Direct vehicle fuel effect
-> Transport services effect
-> Food and groceries effect
-> Utilities effect
-> Annual household impact
```

The crude-to-retail edge references the local retail starting price and selected transmission. The retail-to-direct edge references monthly fuel use. Each indirect edge references category spending and pass-through. Four aggregation edges retain the same source and assumption context as they flow into the annual total.

The reusable React Flow inspector now has an oil-specific layout, unit formatting and formula ledger. Nodes and arrows remain read-only, pointer- and keyboard-selectable, responsive and source-backed.

11. API and Contract

POST /v1/simulations/oil-price accepts the canonical `SimulationRequest` with `scenario.type = oil_price_change` and returns `SimulationResult` model version 1.0.0.

The contract adds `paise_per_litre` and `litres_per_month` assumption units so USD per barrel, retail price, monthly volume and household money cannot be confused. The canonical `oil-price-request.v1.json` fixture is validated against the JSON Schema that generates TypeScript types and defines the strict Pydantic mirror.

The protected Next.js route requires a signed-in session and forwards the request to the server-only engine address. FastAPI rejects missing, incorrectly typed or out-of-range inputs with `INVALID_OIL_PRICE_ASSUMPTION`. The browser never receives the private engine address.

12. Test and Verification Evidence

Check	Result
Economic-engine API and calculation suite	51 passed
Day 10 oil-price engine tests	6 passed
Python lint and formatting	Passed
Canonical contracts and oil-price fixture	Passed
Profile ownership, RLS, money and consent assertions	Passed
Repository secret scan	Passed
ESLint, generated routes and TypeScript	Passed
Next.js production build	Passed; oil page and API generated
Complete desktop and mobile browser suite	31 passed; 9 intentional device skips
Oil result-state accessibility	No serious or critical violations
Mobile preset, graph and horizontal-overflow check	Passed
Sites Vinext production bundle	Passed
Sites production deployment	Version 6 succeeded

The Python suite reports one dependency deprecation warning from FastAPI's Starlette test client regarding `httplib`. It does not affect the 51 passing tests.

13. Production Deployment

The Day 10 progress build is live at ripplelab-progress.nayaksiddhartha397.chatgpt.site. Sites version 6 was built, saved and deployed from exact pushed commit 95fa368. Access remains owner-only during development.

The landing page, progress page, design system and other read-only surfaces are operational. The authenticated oil-price, inflation and repo-rate simulators are complete and verified locally. Hosted account, profile and simulation execution still requires production Supabase values and a hosted FastAPI endpoint; no placeholder credentials were committed.

14. Important Files

- `services/economic-engine/src/ripplelab_engine/oil_price.py` and `test_oil_price.py` - deterministic direct and indirect models plus known-value, direction, exposure, validation and API coverage.
- `apps/web/src/lib/scenarios/oil-price.ts` and `oil-price-simulator.tsx` - profile-derived request, presets, editable form, outputs and channel drivers.
- `apps/web/src/app/api/simulations/oil-price/route.ts` - authenticated server-to-engine boundary.
- `causal-graph-explorer.tsx` and `oil-price.spec.ts` - reusable graph presentation plus desktop, mobile and accessibility checks.
- `oil-price-request.v1.json` and `simulation-contract.schema.json` - canonical fixture and explicit unit vocabulary.
- `apps/web/public/og-day10.png` - verified Day 10 social-preview card.

The social card was generated once with the built-in image tool from an exact-text fintech editorial brief. The inspected 1536 x 1024 PNG contains the requested wording without extra text.

15. Decisions and Limitations

- Keep crude oil and retail fuel as separate inputs; do not pretend the latter moves one-for-one.
- Treat pass-through as editable sensitivity, not an economic forecast.
- Annualize the entered monthly exposure without predicting household behavior changes.
- Exclude exchange-rate, tax, dealer-margin and product-mix calculations in v1.
- Exclude rent changes from the transit-renter preset.
- Do not model fuel substitution, reduced driving or public-policy responses.
- Keep sensitivity deterministic; Monte Carlo remains a later checkpoint.
- Continue treating results as educational simulations rather than financial advice.

No code blocker remains for Day 10.

16. Day 11 Handoff

Day 11 should add the income-tax engine. It should model deterministic changes to annual tax and monthly take-home pay under explicit effective-rate or slab assumptions, preserve India-specific units and evidence, compare several income bands, and return the shared result and causal-graph envelope.

Sign-off: Prepared by Codex | Evidence reviewed: Yes | Ready to continue: Yes